

DODGE CENTER, MN, USA

WMO: 726596

Lat: 44.018N

Lon: 92.831W

Elev: 1305

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04978

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.8	-18.0	2.2	-9.9	-14.7	2.6	-6.8	28.9	21.0	26.7	20.4	8.7	300	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	88.1	74.0	84.2	72.0	81.9	70.2	77.2	83.6	75.2	81.9	73.2	79.5	13.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	138.0	81.5	72.7	127.5	79.6	71.7	123.1	78.6	41.5	83.1	39.7	81.7	37.7	80.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.7	24.1	20.9		-18.5	92.6	5.8	3.1	-22.7	94.8	-26.0	96.6	-29.3	98.4	-33.5	100.6	
				-18.9	80.6	5.7	2.7	-23.0	82.5	-26.3	84.1	-29.5	85.6	-33.7	87.6	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	15.6	17.5	31.8	45.7	57.4	67.5	70.6	68.0	61.9	48.0	34.9	21.1
	DBStd	21.89	12.97	11.80	12.59	10.27	9.06	6.27	5.40	5.54	7.92	9.59	10.94	11.62
	HDD50	4312	1066	911	579	199	32	0	0	0	7	156	465	897
	HDD65	7781	1531	1331	1030	582	269	43	13	32	151	535	903	1361
	CDD50	2534	0	0	14	69	262	526	638	558	363	92	12	0
	CDD65	532	0	0	1	2	35	119	186	125	57	7	0	0
	CDH74	4275	0	0	3	48	355	961	1511	876	454	67	0	0
	CDH80	1083	0	0	0	11	107	272	396	183	103	11	0	0
Wind	WSAvg	9.7	10.9	10.9	10.9	11.7	11.0	8.9	7.3	6.4	8.4	9.6	10.4	10.7
Precipitation	PrecAvg	34.8	1.0	0.8	2.0	3.3	4.0	5.4	4.3	5.0	3.6	2.4	1.9	1.1
	PrecMax	41.4	2.4	1.6	4.1	6.4	7.8	10.3	8.5	12.6	11.8	6.2	6.5	2.7
	PrecMin	23.1	0.4	0.1	0.6	1.0	1.1	2.6	1.1	0.9	1.0	0.6	0.2	0.2
	PrecStd	5.3	0.6	0.4	1.0	1.6	1.7	2.3	2.0	2.4	2.5	1.5	1.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	45.0	71.5	81.2	89.9	90.9	90.6	88.3	88.3	81.5	68.1	52.0
		MCWB	40.0	41.4	59.0	64.2	67.6	73.1	77.5	76.3	71.3	64.8	57.4	46.7
	2%	DB	37.5	39.2	61.4	72.4	82.0	86.4	88.0	84.1	82.3	73.4	61.0	43.3
		MCWB	34.8	36.6	54.6	57.7	63.8	71.4	76.8	74.5	69.5	61.5	53.2	41.0
	5%	DB	36.0	36.6	54.7	66.3	78.6	82.2	84.0	81.6	79.5	69.7	54.9	38.5
		MCWB	34.1	34.7	49.7	54.9	62.8	68.4	74.2	72.1	67.3	59.9	49.1	36.6
	10%	DB	33.8	34.2	48.0	62.9	72.6	79.5	81.6	79.2	75.2	63.3	50.1	36.1
		MCWB	32.6	32.7	44.8	51.8	59.7	66.9	71.9	69.9	65.8	54.6	45.6	34.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.1	41.9	61.5	66.5	73.0	78.0	82.1	78.4	75.3	70.8	59.9	48.4
		MCDB	43.4	43.8	68.4	75.6	81.3	86.1	88.1	83.8	83.1	76.5	65.9	49.8
	2%	WB	36.2	37.7	55.8	60.5	68.2	74.9	78.3	75.9	72.4	64.4	54.7	41.1
		MCDB	38.2	39.5	59.9	68.6	76.7	82.6	84.0	82.3	79.4	71.4	59.2	43.8
	5%	WB	34.0	35.1	50.0	56.8	65.7	72.4	75.9	74.0	70.0	59.5	50.1	37.1
		MCDB	35.3	36.4	55.0	64.4	73.5	79.2	82.2	79.9	76.9	65.7	55.5	38.3
	10%	WB	31.8	32.4	44.7	53.4	62.7	69.9	73.7	71.8	67.4	55.7	45.7	34.3
		MCDB	33.3	34.0	48.1	60.7	69.5	76.6	79.9	77.4	74.3	62.9	49.1	35.5
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.5	15.2	18.7	19.4	18.2	18.3	18.5	20.1	18.3	15.5	13.2
		MCDBR	16.3	15.8	22.7	27.2	25.5	22.4	19.9	19.9	23.2	25.7	22.7	15.7
		MCWBR	14.3	14.0	16.1	16.3	12.1	11.0	11.3	11.8	12.5	14.2	15.4	13.8
	5% WB	MCDBR	15.1	15.1	21.4	23.9	20.7	18.8	18.5	18.0	19.7	20.6	21.6	14.4
		MCWBR	13.5	13.8	16.0	16.3	12.4	11.9	12.1	11.9	12.3	14.6	15.7	13.0
Clear-Sky Solar Irradiance	taub	0.263	0.289	0.313	0.364	0.391	0.402	0.401	0.397	0.366	0.321	0.287	0.263	
	taud	2.425	2.338	2.405	2.316	2.293	2.303	2.303	2.322	2.381	2.497	2.510	2.451	
	Ebn at Noon	279	287	294	285	279	276	274	272	272	273	268	267	
	Edh at Noon	23	30	32	39	41	41	40	38	33	26	21	20	
All-Sky Solar Radiation	RadAvg	557	895	1189	1454	1704	1902	1997	1682	1334	861	561	409	
	RadStd	62	95	117	154	133	148	82	115	95	116	51	36	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page